

Trend Analysis of Global Temperature From 1922 to 2022;- A case study Report

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Abstract

Currently increasing global surface temperature is one of the major concerns of humanity. Several studies have recently investigated with increasing accuracy time series of globally averaged observations, available since the last century. Overall, there is widespread agreement that global Average temperature has been rising over the 20th century and high confidence that such a trend is accelerating. Trend analysis of temperature data obtained from HadCURT over the entire globe is done for a period of 1922 to 2022. An alarming increasing trend in the temperature, which is 0.012 degrees celsius per year is observed in the dataset. Linear regression model predicted a temperature value of 19.26 ± 0.409 for the year 2072 which is very high compared to the current value which is 18.73 ± 0.405 degree celsius.

Keywords: Temperature, HadCURT, Linear regression

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1. Introduction

The Earth's climate has been changing continuously for millions of years, but the recent trend of global temperature rise has caused growing concern Masson-Delmotte et al. (2018). Climate change, driven by human activities such as burning fossil fuels, has increased the concentration of greenhouse gases in the atmosphere. These gases trap heat and cause the

Earth's temperature to rise, resulting in various environmental impacts such as sea-level rise, extreme weather events, and loss of biodiversity.

Scientists have been collecting and analyzing temperature data from around the world better to understand the extent and nature of global temperature rise. The temperature data is provided by NOAA, Met Office, IMD, CPC and many other leading organisations available, and the world unanimously agrees on an increasing temperature after the industrial revolution. The greenhouse gas effect and the associated temperature have been increasing exponentially in the Anthropocene era Ruddiman (2013).

But, This warming trend has not been uniform over time, and there have been periods of more rapid warming and slower warming. The most recent decade (2006-2015) was the warmest on record, with an average temperature in the Northern Hemisphere of about 0.8°C (1.4°F) above the 1981-2010 average. This can have implications for variations in temperature anomaly. In this project, we aim to conduct a trend analysis of global temperature from 1922 to 2022 to understand the changes happening in the past 100 years. By analyzing long-term trends, we can gain a deeper understanding of the nature and magnitude of climate change and its potential impact on the environment and human societies.

2. Materials and Methods

The time series analysis in this study is based on linear multiple regression models. Residual analysis is therefore necessary in order to verify the homoscedasticity and independence of the residuals since they are predicated on the assumption that the residuals of regression models are independent and identically distributed, with mean 0 and finite variance. A best linear

unbiased estimator is then provided for the linear multiple regression models by modelling residual correlation using autoregressive stationary processes. Lastly, in cases when the distribution of the residuals is unknown, the bootstrapping technique is employed to estimate the parameters.

2.1. Data

The global annual temperature data was obtained from The Met Office Hadley Centre. The HadCURT4.6.0 near surface temperature data set is produced by blending data from the CRUTEM4 surface air temperature dataset and the HadSST3 sea-surface temperature. dataset (Morice et al., 2012) The data is provided on a grid of boxes covering the globe, with values provided for only those boxes containing temperature observations in a particular month and year. Interpolation is not applied to infill missing values. The first version of HadCRUT initially spanned the period 1881–1993, and this was later extended to begin in 1850 and to be regularly updated to the current year/month in near real-time. The mean temperature is considered to be 18 degrees celsius for the simplicity of the analysis.

2.2. Linear Regression

When modelling the relationship between a scalar response and one or more explanatory variables in statistics, linear regression is a linear method (also known as dependent and independent variables). Simple linear regression is used when there is only one explanatory variable, and multiple linear regression is used when there are numerous variables Cohen et al. (2014). The basic linear regression equation can be written as:

$$\vec{y} = \hat{a}\vec{x} + \hat{b} + \vec{\epsilon} \quad (1)$$

Here y is the dependent variable (Temperature) and x is the independent variable, which is the time (Years).

$$\hat{a} = \frac{\sum_i x_i y_i - \frac{1}{n} \sum_i x_i \sum_i y_i}{\sum_i x_i^2 - \frac{1}{n} (\sum_i x_i)^2} \quad (2)$$

$$\hat{b} = \frac{1}{n} (\sum_i y_i - \hat{a} \sum_i x_i) \quad (3)$$

Here, a crucial presumption is that the error term is made up of normally distributed data points. An independent and identically (iid) Gaussian distribution is said to be comprised of such data. To confirm that mistakes are unrelated, this is done. Overall, linear regression analysis provided us with valuable insights into the long-term trend of global temperature, and helped us to identify any significant changes or patterns in the data over the past century.

2.2.1. Pearson's correlation coefficient

A measure of linear correlation between two sets of data is the Pearson correlation coefficient (PCC), also known as Pearson's r, the bivariate correlation, or simply the correlation coefficient. It is essentially a normalized measurement of the covariance, with the result always falling between 1 and -1. It is the

ratio between the covariance of two variables and the product of their standard deviations. The value of correlation coefficient, 'r' can be determined using the following formula.

$$r = \frac{\sum_{i=1}^n (x_i - \frac{1}{n} \sum_{j=1}^n x_j)(y_i - \frac{1}{n} \sum_{j=1}^n y_j)}{\sqrt{\sum_{i=1}^n (x_i - \frac{1}{n} \sum_{j=1}^n x_j)^2} \sqrt{\sum_{i=1}^n (y_i - \frac{1}{n} \sum_{j=1}^n y_j)^2}} \quad (4)$$

An r value of 1 means that both variables have a perfect linear relationship, where an increase in one variable will result in an increase in the second variable as well. An r value of -1 means that both datasets are having a perfect correlation but in opposite directions, as an increase in one variable will result in a decrease in the second variable.

2.3. Residual Analysis

To evaluate the quality of the linear regression model, we conducted a residual analysis by examining the differences between the observed values and the predicted values of the dependent variable. We calculated the residuals for each observation and plotted them against the predicted values to assess the pattern and variability of the residuals. The residuals were normally distributed around zero, indicating that the model assumptions of normality and linearity were satisfied. The scatter plot of the residuals also showed no discernible pattern or trend, indicating that the model assumption of homoscedasticity was satisfied (Wetherill, 2012).

We first plotted the residuals against the predicted values of the dependent variable to visually assess whether the residuals were randomly distributed around zero. We also examined the residual plot for any patterns or trends, such as a funnel shape or increasing or decreasing variance, which would indicate that the model assumptions were violated.

2.3.1. Pearson's χ^2 -test

The χ^2 -test is intended to test how likely it is that an observed distribution is due to chance. It is also called a "goodness of fit" statistic because it measures how well the observed distribution of data fits with the distribution that is expected if the variables are independent. Here The null hypothesis that the residuals are normally distributed as $N(0, \sigma_e^2)$ can be tested with this test.

$$c^2 = \sum_{i=1}^m \frac{(r_i - d_i)^2}{d_i} \sim \chi_{m-1}^2 \quad (5)$$

2.3.2. Durbin-Watson Test

The Durbin Watson (DW) statistic is a test for autocorrelation in the residuals from a statistical model or regression analysis. One of the premises of the linear regression model is broken by autocorrelation, which occurs when the residuals of a regression model are not independent of one another. the Durbin-Watson statistic d, where,

$$d = \frac{\sum_{i=2}^n (r_i - r_{i-1})^2}{\sum_{i=1}^n r_i^2} \quad (6)$$

The Durbin-Watson statistic will always have a value ranging between 0 and 4. A value of 2.0 indicates there is no autocorrelation detected in the sample. Values from 0 to less than 2

points to positive autocorrelation and values from 2 to 4 means negative autocorrelation.

2.4. Bootstrapping

Bootstrapping estimates the properties of an estimand (such as its variance) by measuring those properties when sampling from an approximating distribution Tibshirani and Efron (1993)Efron (2003) The residuals must be independent and identically distributed with a general distribution with a mean of 0 and a finite variance in order to execute bootstrapping. In this case, it is impossible to make any assumptions about the parameters' distributional form, hence the analytical discussion from earlier cannot be used to calculate confidence ranges for parameters. If the Durbin-Watson test is successful, but Pearson's χ^2 -test is not, then this is the default technique used to compute the parameters and confidence intervals. If the residuals are not independent, the aforementioned method fails to get the appropriate estimates of the parameters and their confidence intervals because resampling always eliminates correlation.

2.4.1. Confidence Interval

A confidence interval is the mean of your estimate plus and minus the variation in that estimate. This is the range of values you expect your estimate to fall between if you redo your test, within a certain level of confidence.

2.5. Programming language

Python packages such as numpy, pandas, ingress, stats.model and scipy for the statistical analysis and seaborn and matplotlib for the visualisation of data and results.

3. Result and Discussion

As mentioned earlier, The HadCURT Annually averaged Temperature data is used from 1922 to 2022 to identify the correlation between temperature and time in the past century. The raw data is plotted with respect to time in Figure:1 The time series shows an increasing trend in the temperature value during the study period. The red line indicates the 5-year running mean of temperature, which reduces the fluctuation in the data and helps us to understand the trend of the data. The Time series data shows a significant global temperature increase during the time of the second world war which went on between 1939 to 1945. The second world war was known for the usage of modern war machines and nuclear warfare. The technological advancement that each country displayed in the war led to an exponential increase in greenhouse gas concentration in the atmosphere and associated temperature increases.

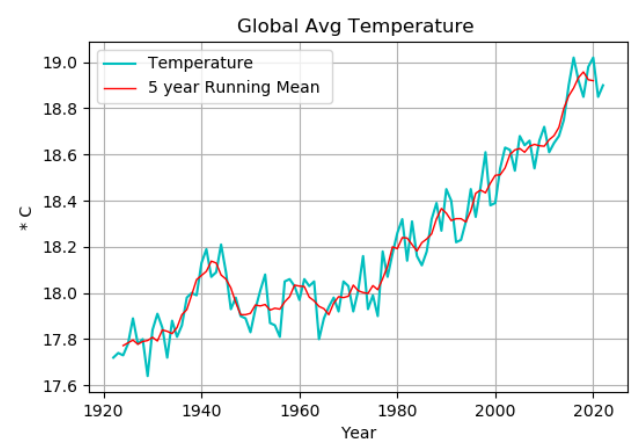


Figure 1: The time series of Global annual mean temperature from 1922 to 2022

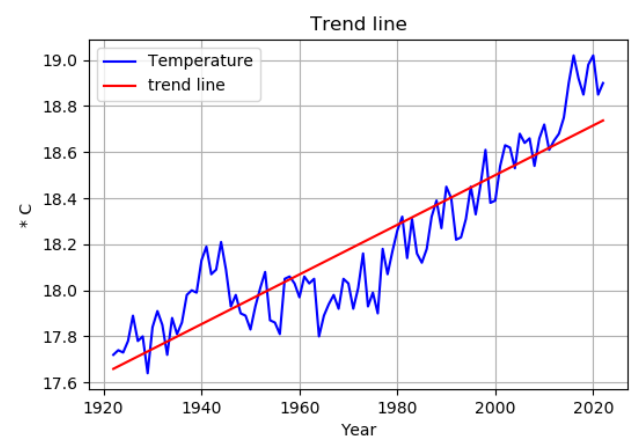


Figure 2: Trendline created using linear model represented using red solid line

3.1. Linear Regression model

Apart from the unfortunate increase, the temperature had a relatively constant increase with time after1950s. A linear regression was performed over the data. The r value obtained was 0.90 which shows a significant positive correlation between the variables. The trend line observed is plotted in Figure :2. The corresponding **t-value** for the model is **20.67** The *t*-value is greater than $t_{crit}(1 - \frac{\alpha}{2}, N - 2)$ for a significance level $\alpha = 0.05$ and 101 points. The null hypothesis of an absence of a trend in the temperature is hence rejected. The r-value suggests a significant positive correlation between the variables. The confidence interval of the estimated data is plotted in Figure:3.

3.2. Treating residuals

Error analysis is performed in order to test the hypothesis of identically and independently Gaussian-distributed residuals which is shown in Figure:5. The frequency distribution of the residuals are shown in the histogram in Figure:4.

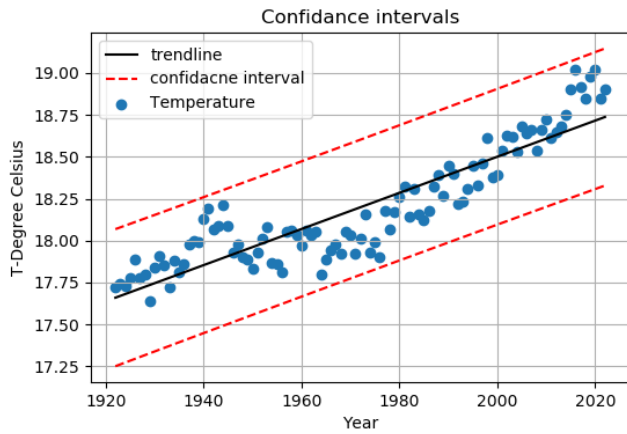


Figure 3: The confidence interval for the obtained dataset represented by red dotted line

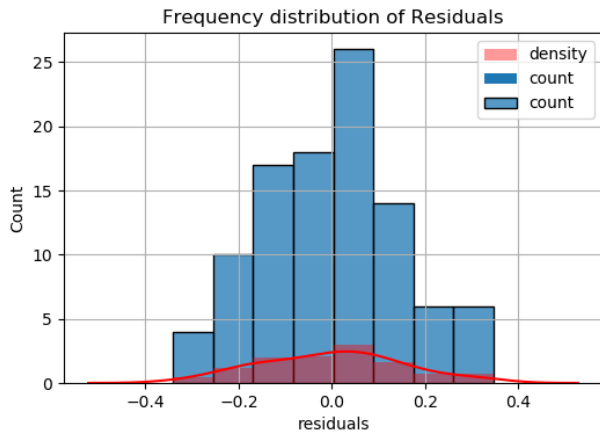


Figure 4: The frequency distribution of residues, while the distplot on the bottom(red) represents the frequency distribution and histogram represents the same but in terms of count of incidents

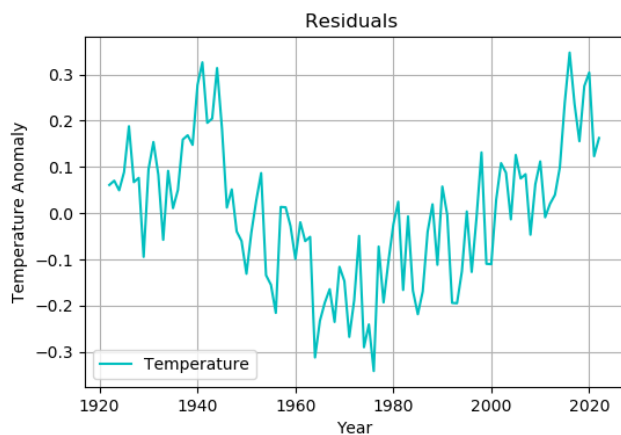


Figure 5: Time series plot for residuals

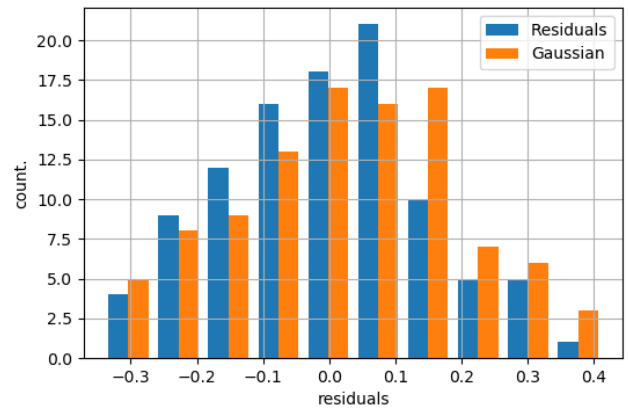


Figure 6: Histogram of residuals for the linear model compared with tested Gaussian distribution

The trend suggests that between 1960 and 2000 there was a higher frequency of lower temperature anomalies. A decreasing trend can be observed after the second world war while it started to increase in the current century again. After 2000, the frequency of higher temperature anomaly has increased again to a noticeable level.

3.3. Testing Gaussian Distribution

The histogram of residuals (Figure:4) is compared with a random gaussian/normal distribution to test the null hypothesis that states the residuals are distributed in a normal distribution. Performed χ^2 -test test with the null hypothesis that residuals are distributed according to Gaussian distribution returns a value of $c2 = 14.82$. The null hypothesis of Gaussian distributed residuals is therefore rejected with a 99 % confidence level.

3.4. Testing for the independence of residues

The Durbin-Watson statistics parameter for the residues was computed as 0.68. The small Durbin-Watson test statistic (j_1) signifies the presence of strong positive autocorrelation in the residuals, which indicates that the model may not be a good fit for the data.

3.5. Bootstrapping

The data was predicted for a period of 10,20, and 50 through bootstrapping. The prediction for 50 years is plotted in Figure: while for the observed data, the temperature predicted is 18.73 ± 0.405 degrees Celsius. For 50 years, 30 years and 20 years the predicted values are 19.26 ± 0.409 , 19.05 ± 0.407 and 18.94 ± 0.406 degrees Celsius in order.

The future trends predict a positive temperature anomaly value, which suggests that there is a higher chance for the year to be warmer than expected.

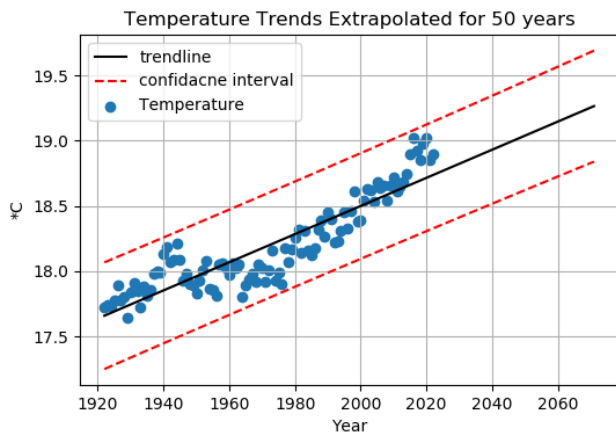


Figure 7: Bootstrap graph for the linear trend for 50 years.

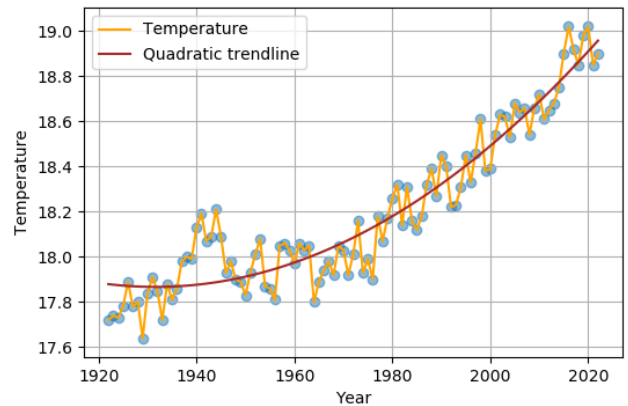


Figure A.8: Second degree trendline for HadCURT Temperature data over the last 101 years

4. Conclusions

The analysis shows that the HadCURT temperature data shows a huge positive correlation in the past 100 years. As per statistical models, the temperature is going to rise in the future at a rate of 0.012 degrees celsius per year in the upcoming future. Sudden actions to control climate change have to be done in order to put an end to the alarming rise in temperature. The study also shows the effects of world war two in the atmospheric temperature rise as well.

Appendix A. Appendix trying Quadratic fits

An additional attempt to do the quadratic regression over the temperature data was done. But the prediction parts were not completed in time, still, the obtained results are discussed here.

Appendix A.1. Quadratic regression model

A second-degree quadratic regression fit was done over the data and is represented in the figure:A.8. The plot shows a perfect fit compared to linear regression model outputs. The frequency distribution of new residuals is done and are represented in Figure:A.9 and the normal time series analysis of the residual data is shown in Figure:A.10. As we can observe from the plots, the residuals have decreased significantly compared to linear regression model outputs.

Appendix A.1.1. Durbin-Watson test

The Durbin-Watson statistics parameter for the residues was computed as 1.13. which is comparatively better than the results obtained through linear regression analysis. The new Durbin-Watson test statistic signifies the presence of strong positive autocorrelation in the residuals, which indicates that the model may be a good fit .for the data

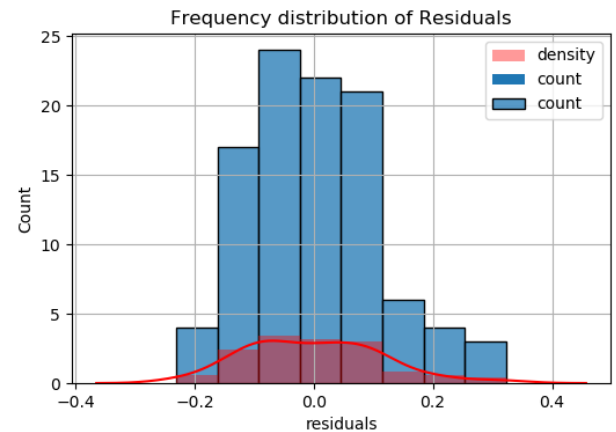


Figure A.9: The frequency distribution of residues, while the distplot on the bottom(red) represents the frequency distribution and histogram represents the same but in terms of count of incidents

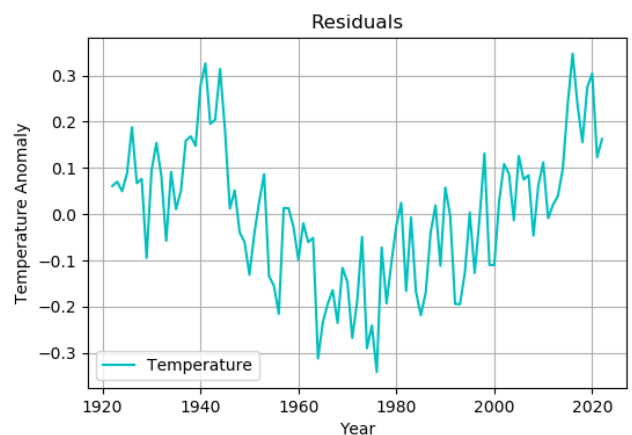


Figure A.10: time series plot for residual

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